Leuprorelin (Prostap®) for breast cancer

Leuprorelin is a cancer drug. It is used to treat breast cancer and prostate cancer.

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What is leuprorelin (Prostap®)?

Leuprorelin is a hormonal therapy drug. It is also called Prostap®. Leuprorelin is used to treat breast cancer.

It is best to read this information with our general information about hormonal therapy and the type of breast cancer you have.

Leuprorelin can also be used to treat prostate cancer. We have separate information about leuprorelin for prostate cancer.

Your cancer team will talk to you about this treatment and its possible side effects before you agree (consent) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

How leuprorelin works

Hormones are chemicals that our bodies make. They act as messengers and help control how cells and organs work. Hormonal therapy drugs change how hormones are made or work in the body.

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Oestrogen can encourage breast cancer to grow. Leuprorelin stops the ovaries making oestrogen and lowers the level of oestrogen in the body. You can only have leuprorelin if you have not been through menopause. You may have it with other hormonal therapy drugs or on its own.

How leuprorelin is given

You have leuprorelin as an injection under the skin (subcutaneously) or into a muscle (intramuscular).

During your treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

You usually have leuprorelin once every 3 months. Your nurse or doctor will talk to you about your treatment plan.

You have the injections at a GP surgery. A practice nurse gives them to you. If you are not able to visit the surgery, a district nurse can give you the injections at home.

If you are taking any medicines to thin your blood, tell your doctor, nurse or pharmacist. These medicines can make you bruise more easily.

Your cancer team will discuss your treatment plan with you. They may give you other medicines to take home. Take all your medicines exactly as they tell you to. Do not stop taking any of your medicines unless your doctor tells you to.

Injection site

The skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker than the surrounding skin, if you have black or brown skin.

If this happens, let your doctor know. Painkillers may help.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

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Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

When you have this treatment, you will have regular appointments with a doctor, nurse or pharmacist. Always tell them about any side effects you have. They can give you:

- drugs to help control most side effects
- advice about managing side effects.

Most side effects can be managed. But sometimes side effects are harder to control. It is important not to stop taking hormonal therapy without telling your doctor. If side effects cannot be managed, your doctor may suggest a different type of hormonal therapy.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Hot flushes and sweats

Hot flushes are a common side effect of this treatment. During a flush, your neck and face may feel warm. If you have white skin, your skin may look red. If you have black or brown skin, your skin colour may deepen. Flushes may last up to 5 minutes. You might also have sweats or feel anxious or irritable during a hot flush.

Here are some things you can do to reduce the effects of hot flushes:

- Wear natural fabrics, such as cotton.
- Wear layers so you can remove clothes as needed.
- Use cotton sheets and layers of bedding you can remove.
- Try cooling pads or pillows to keep you cool.
- Keep rooms cool or use a fan.
- Have cold drinks and avoid caffeine and alcohol.

You might have fewer hot flushes as your body adjusts to this treatment. You could try therapies to help you cope with hot flushes, such as:

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- talking therapies like cognitive behaviour therapy (CBT)
- controlled breathing exercises
- yoga
- acupuncture.

If they do not improve, talk to your doctor. Certain drugs can help to improve hot flushes.

Flushes usually stop a few months after treatment ends. But some people continue to have them.

Difficulty sleeping

You may have difficulty sleeping while taking this treatment. We have more information about managing sleep problems. Tell your doctor or nurse if it does not get better.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Bone pain

Some people may get pain in their bones. Tell your doctor or nurse if this happens.

Muscle or joint pain

You may get pain in your muscles or joints. If this happens, tell your doctor or nurse. They can give you advice and painkillers or other drugs to help. Being physically active and keeping to a healthy weight may help with the pain. It can keep your joints flexible and reduce stress on them. If the pain does not get better, tell your doctor or nurse. They may talk to you about changing to a different hormonal drug.

Raised blood sugar levels

This treatment can raise your blood sugar levels and increase the risk of diabetes.

Signs of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.

Tell your doctor or nurse if you have these symptoms. They can do tests to check your blood sugar level and give you advice.

If you already have diabetes, your blood sugars may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

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Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Feeling tired (fatigue)

Feeling tired is a common side effect. Try to pace yourself and plan your day so you have time to rest between activities.

Being physically active can help to manage tiredness and give you more energy. It also:

- helps you sleep better
- reduces stress
- improves your bone health.

If you feel sleepy, do not drive or operate machinery.

Muscle weakness

You might notice that your muscles feel weaker than usual. Tell your doctor or nurse if this happens.

Feeling sick

Leuprorelin can make you feel sick (nausea) but this is usually mild. If it does not improve, your doctor can give you anti-sickness drugs to help.

Diarrhoea or constipation

Leuprorelin may cause some bowel changes, but this is not usually severe. If this is a problem tell your doctor or nurse.

Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. If you have a stoma, it may be more active than usual. Drink enough fluids to keep you hydrated. Your doctor may advise you to take anti-diarrhoea medicines.

Constipation means that you are not able to pass stools as often as you normally do. Drinking plenty of fluids and eating high-fibre foods, such as fruit, vegetables and wholemeal bread, can help. Your doctor may advise you to take drugs called laxatives if needed.

Weight gain

You may gain weight when you are having this treatment. Eating healthily and being active can help you keep to a healthy weight. Your doctor, nurse or dietitian can give you more advice.

Skin changes

This treatment can affect your skin. It may cause a rash, which may be itchy. Your doctor or nurse can tell you what to expect. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

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Always tell your doctor or nurse about any skin changes. They can give you advice and may prescribe creams or medicines to help. Changes to your skin usually improve when treatment ends.

Serious skin changes

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

Hair thinning

While you are having this treatment, your hair may become thinner, drier and more brittle. This is usually mild. If you are worried about this, ask your nurse for advice on hair care.

Effects on the eyes

If you notice any change in your vision it is important to tell your doctor or nurse.

Dizziness

Leuprorelin may make you feel dizzy. Tell your doctor or nurse if this happens. If you feel dizzy do not drive or operate machinery.

Mood changes

You may feel low or depressed, or have mood swings, during this treatment. Talking to family and friends about how you feel might help. If mood changes last for more than a few weeks, tell your doctor, nurse or pharmacist. They can talk to you about different ways to manage low mood or depression.

Memory and concentration

You may notice changes in your memory. You may also find it harder to concentrate.

To help you remember things, try using:

- reminder alarms or notes on your phone
- a diary, calendar or phone app
- a notebook you carry with you.

Breast swelling or tenderness

This treatment may cause swelling and tenderness of your breast tissue.

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Changes to periods

If you have periods, these will usually stop while you are having this treatment. If this does not happen, tell your doctor.

Periods might return after treatment, but this does not always happen. Your menopause may start sooner than it would have done. Your doctor, nurse or pharmacist can give you more information.

You can still become pregnant even if your periods have stopped or become irregular. This means you will need to use effective contraception. You can discuss this with your doctor or nurse.

Vaginal dryness or bleeding

This treatment can cause vaginal dryness. Non-hormonal creams, gels or lubricants can help with vaginal dryness and any discomfort during sex. You can buy these from a chemist, or your doctor can prescribe them.

You might have slight vaginal bleeding for a short while:

- when you start treatment
- if you change from 1 hormonal treatment to another.

Vaginal dryness can also cause bleeding. Tell your doctor or nurse if you have any vaginal bleeding, especially if it lasts for more than a few days.

Bone thinning (osteoporosis)

Taking this treatment for several months or more can increase the risk of bone thinning. This is called osteoporosis. It can make you more likely to get a broken bone (fracture).

You may have bone density scans to check your bone health before and during treatment. These scans can diagnose osteoporosis or osteopenia (low bone density) so that it can be monitored. If your scan shows osteoporosis, your doctor usually prescribes:

- drugs to protect your bones called bisphosphonates
- calcium and vitamin D supplements.

Regular exercise where you support your body weight, like walking, running and dancing, can help look after your bones. This is called weight-bearing exercise. Other types of strength exercises are also helpful. If you have any bone thinning or the cancer is affecting your bones, get advice from your cancer team first.

Eating a healthy diet and not smoking also helps take care of your bones.

Loss of appetite

This treatment can affect your appetite. Don’t worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

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Tingling in the hands or feet

Leuprorelin may cause tingling in your hands or feet but this is not common. If this happens tell your doctor or nurse.

Effects on the liver

This treatment may affect how your liver works. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your liver is working.

Effects on the heart

This treatment might increase certain risk factors for heart problems or cause:

- fast heartbeat (palpitations)
- high blood pressure
- high levels of cholesterol – a fatty substance in your blood.

Talk to your doctor if you are worried about these. They can monitor you or treat you if needed.

Things you can do to take care of your heart include:

- not smoking
- maintaining a healthy weight
- being physically active.

Raised blood sugar levels

This treatment can raise your blood sugar levels and increase the risk of diabetes.

Signs of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.

Tell your doctor or nurse if you have these symptoms. They can do tests to check your blood sugar level and give you advice.

If you already have diabetes, your blood sugars may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a blood clot. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

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- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

Contraception

Your doctor, nurse or pharmacist will advise you not to get pregnant while having this treatment. The drug may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use, for example using barrier contraception instead of hormonal contraception like the oral pill
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can get pregnant or make someone pregnant. If you are worried about this, it is important to talk with your doctor before you start treatment.

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Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

Author

Date reviewed

Reviewed: 01 January 2025 | Next review: 01 January 2027

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
-

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use illustrations to explain text

- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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